

CISCO CERTIFIED CYBEROPS ASSOCIATE (CBROPS 200-201)

Lab 5.3.7: Introduction to Wireshark

Mininet Network Virtualization, PDU Layer Encapsulation, Local vs. Remote ICMP Traffic Analysis

COURSE MODULE

Module 5: Network Protocols

CORE PROTOCOLS

Ethernet II, IPv4, ICMP (RFC 792)

SIMULATION ENGINE

Mininet SDN Emulator (OpenFlow)

WORKSTATION ENVIRONMENT

CyberOps Workstation VM

Objectives

- **Part 1: Install and Verify the Mininet Topology** — Execute Python topology orchestration scripts (`cyberops_topo.py`), launch host terminals (`xterm`), and baseline IP/MAC configurations.
- **Part 2: Capture and Analyze ICMP Data in Wireshark** — Capture local LAN ping sequences (H1 → H2), dissect 3-pane Wireshark structures, and analyze remote subnet packet routing (H1 → H4 via Router R1).
- **Protocol Encapsulation Analysis** — Correlate how Protocol Data Units (PDUs) transition down the OSI stack from Application payload to Layer 2 Ethernet frames.

Background / Scenario

Wireshark is the industry-standard software packet analyzer and protocol dissection tool used by Security Operations Center (SOC) analysts, network engineers, and forensic investigators. Wireshark captures network frames traversing physical or virtual interfaces and decodes their protocol layers according to RFC specifications.

In this lab, the Mininet software-defined network emulator is utilized within the CyberOps VM to construct a multi-node virtual topology containing hosts (H1, H2, H3, H4), an OpenFlow switch (S1), and a perimeter router (R1). Analysts observe and compare packet encapsulation during local LAN communication versus inter-network routed communication.

Part 1: Install and Verify the Mininet Topology

Step 1 & 2: Launch Topology Orchestration Script

```
[analyst@secOps ~]$ sudo ~/lab.support.files/scripts/cyberops_topo.py
[sudo] password for analyst: cyberops
*** Creating network
*** Adding controller
*** Adding hosts:
H1 H2 H3 H4 R1
*** Adding switches:
s1
*** Adding links:
(H1, s1) (H2, s1) (H3, s1) (s1, R1) (R1, H4)
*** Starting CLI:
mininet>
```

Step 3: Host Addressing Baseline Configuration Table

Launching host terminals: `mininet> xterm H1 H2` and executing `ip address` on each node:

HOST NODE & INTERFACE	IPV4 ADDRESS / SUBNET	MAC ADDRESS (LAYER 2)	SUBNET PLACEMENT & ROLE
H1 (H1-eth0)	10.0.0.11/24	ba:d4:1d:7b:f3:61 (or simulated MAC)	Local LAN Host (Initiating endpoint).
H2 (H2-eth0)	10.0.0.12/24	ba:d4:1d:7b:f3:62 (or simulated MAC)	Local LAN Host (Same broadcast domain).
H3 (H3-eth0)	10.0.0.13/24	5a:d0:1d:01:9f:be	Local LAN Host.
R1 (R1-eth1)	10.0.0.1/24	00:00:00:00:00:01	Default Gateway Router Interface for LAN 10.0.0.0/24 .
R1 (R1-eth2)	172.16.0.1/24	00:00:00:00:00:02	Gateway Interface for Remote Subnet 172.16.0.0/24 .
H4 (H4-eth0)	172.16.0.40/24	ba:d4:1d:7b:f3:64	Remote Simulated Enterprise Server.

Part 2: Capture and Analyze ICMP Data in Wireshark

Wireshark Three-Pane Architecture

- **Packet List Pane (Top):** Summarizes all captured frames chronologically (Frame #, Time, Source, Destination, Protocol, Length, Info).
- **Packet Details Pane (Middle):** Displays hierarchical protocol encapsulation breakdown (Frame → Ethernet II → IPv4 → ICMP).
- **Packet Bytes Pane (Bottom):** Displays raw payload data in synchronized hexadecimal and ASCII representation.

Step 1: Local LAN Traffic Analysis (H1 Ping → H2 10.0.0.12)

Wireshark is started on node H1 (`wireshark &`) listening on interface `H1-eth0` . Five ping packets are sent: `ping -c 5 10.0.0.12` . Applying display filter `icmp` isolates the ICMP Echo Requests and Replies.

STEP 1.G — LOCAL SUBNET FRAME HEADER VERIFICATION

1. Does the Source MAC address match H1's interface?
2. Does the Destination MAC address in Wireshark match H2's MAC address?

DISSECTION FINDINGS:

- **1. Source MAC Match: Yes.** The Source MAC in the Ethernet II frame matches the physical hardware address of `H1-eth0` (`ba:d4:1d:7b:f3:61`).
- **2. Destination MAC Match: Yes.** The Destination MAC in the Ethernet II frame matches the exact physical MAC address of host H2 (`ba:d4:1d:7b:f3:62`).

Networking Insight: Because host H1 and host H2 reside on the same local Layer 3 subnet (`10.0.0.0/24`), H1 resolves H2's MAC directly using ARP. Frames are switched directly across Switch S1 without requiring routing by Default Gateway R1.

```
# Protocol Data Unit (PDU) Encapsulation Stack:
+-----+
| Layer 2: Ethernet II Header (Src: H1 MAC → Dst: H2 MAC) |
+-----+
| Layer 3: IPv4 Header (Src: 10.0.0.11 → Dst: 10.0.0.12) |
+-----+
| Layer 4: ICMP Echo Request (Type: 8, Code: 0, Seq: 1) |
+-----+
| Payload Data (Timestamp & Random Diagnostic Padding) |
+-----+
```

Step 2: Remote Subnet Traffic Analysis (H1 Ping → H4 172.16.0.40)

Terminals for remote host H4 and router R1 are launched (`mininet> xterm H4 R1`). A new Wireshark capture is started on `H1-eth0` and H1 pings the remote server: `ping -c 5 172.16.0.40` .

STEP 2.E — REMOTE ROUTING ADDRESSING DISSECTION

Review the captured data in Wireshark. List the destination IP and MAC addresses for the ping to H4.

DISSECTION RESULTS:

- **Destination IPv4 Address:** `172.16.0.40` (Remote Server Host H4).
- **Destination MAC Address:** `00:00:00:00:00:01` (MAC address of router interface `R1-eth1` / Default Gateway `10.0.0.1`).

Core CyberOps Analysis:

Host H1 recognizes that destination `172.16.0.40` is on a remote subnet outside of `10.0.0.0/24` . Therefore, while the Layer 3 IP header maintains the end-to-end destination IP (`172.16.0.40`), the Layer 2 Ethernet frame is addressed to the **Default Gateway's MAC address** (`R1-eth1`). Router R1 decapsulates the Layer 2 frame, routes the packet out interface `R1-eth2` , and re-encapsulates it with H4's MAC address.

Step 3: Mininet Graceful Shutdown & Environment Cleanup

```
# 1. Exit Mininet CLI:
mininet> quit

# 2. Execute Mininet Deep Process Cleanup:
[analyst@secOps ~]$ sudo mn -c
[sudo] password for analyst: cyberops
*** Removing excess controllers/datapaths/pings
*** Removing junk from /tmp
*** Removing OVS datapaths
*** Killing stale mininet node processes
*** Cleanup complete.
```

CyberOps Protocol Analysis Summary

- **Local Switching** On a local LAN, Destination MAC matches the target host's physical NIC.
- **Remote Routing** When communicating across subnets, Destination MAC points to the **Default Gateway router**, while Destination IP remains the remote destination host.
- **PDU Encapsulation** Upper-layer protocols (ICMP/TCP/UDP) are encapsulated within IPv4 packets, which are further encapsulated inside Ethernet II frames.
- **Process Cleanup** Always run `sudo mn -c` after Mininet sessions to prevent orphaned OpenFlow switch datapaths and virtual interface leaks.